



Strend

AI-Powered Stock Management & Analysis



"Smart Insights. Smarter Decisions."

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 Meet Jethva

FINTech

AI & LLM

ANALYSIS

Agenda

Presentation Overview



Introduction



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Objectives



Existing System & Challenges



Proposed Solution



Technology Stack



Key Modules & Features



How It Works (Workflow)



AI Agent — The Brain



Results & Achievements



Testing



Challenges Faced



Future Scope



Conclusion



Q&A

Introduction to Strend

What is Strend?



Unified Analysis Platform

Full-stack app combining real-time data fetching, technical indicators (MACD, RSI), and fundamental metrics (P/E, ROE).



AI-Powered Insights

Intelligent chatbot + AI-driven news sentiment analysis via conversational interface.

CORE CAPABILITIES

Technical Analysis

Fundamentals

News Sentiment

Real-time Data

Who is it for?

- ✓ Individual Retail Investors
- ✓ Stock Market Beginners
- ✓ Day Traders & Swing Traders

🎯 **Domain Focus**
FinTech (Financial Technology)

The Value Proposition

Automates manual research → faster data-driven decisions.

RESEARCH TIME

Hours



WITH STREND

Seconds

Problem Statement



The Problem

What needs solving?

- ✕ **Fragmented Research:** Investors spend hours manually checking multiple sites (Screener, MoneyControl, Google News).
- ✕ **Disjointed Tools:** No single platform unifies technical, fundamental, and news analysis with AI.
- ✕ **Complexity:** Beginners struggle to interpret complex financial indicators without guidance.
- ✕ **Manual Tracking:** Hard to track watchlists and compare performance against benchmarks like Nifty.



Target Audience

Who is affected?

- 👤 **Retail Investors:** Individuals managing their own portfolios who need data-driven confidence.
- 🌱 **Market Beginners:** New entrants seeking simplified analysis and educational guidance.
- 📈 **Active Traders:** Day & Swing traders requiring quick, consolidated insights to act fast.



Impact & Urgency

Why does it matter?

- 🔴 **Financial Losses:** Incomplete research leads to bad decisions and avoidable capital loss.
- 🕒 **Productivity Loss:** Significant time is wasted on repetitive data gathering tasks.
- 🔴 **Analysis Paralysis:** Information overload without smart filtering causes confusion and missed opportunities.

Project Objectives



AI & Automation

Core intelligence



Unified Analysis

Automate stock research by combining technical, fundamental, and news analysis in one platform.



Conversational AI

Build an AI chatbot that understands natural language and executes analysis tools on behalf of the user.



Advanced Tools

Key capabilities



Real-Time Watchlist

Provide real-time watchlist management with live price tracking and Nifty comparison.



Automated Reports

Auto-generate daily/weekly stock prediction reports with PDF creation and email delivery.



Visual Analysis

Deliver automated candlestick chart capture and visual analysis directly to users.



User Value

Growth & Efficiency



Learning Journal

Enable a "Learning Journal" where users log trading lessons, sentiments, and exceptions.



Efficiency Boost

Reduce manual stock research effort from hours to seconds through intelligent automation.

Existing System & Challenges

How It's Done Today



Fragmented Ecosystem

4-5 different websites: Screener.in, TradingView, MoneyControl, Chartink.



Manual Workflow

Separate logins, manual copy-pasting of data across tools.



No AI Assistance

No intelligent assistant that maintains context or runs analysis tools conversationally.

TOOLS CURRENTLY JUGGLED

Screener.in

TradingView

MoneyControl

Chartink

Google News

Current Limitations



No Unified Platform

Technical, fundamental, news analysis in isolated silos.



High Learning Curve

No natural language interface; complex UI per tool.



Missing Feedback Loop

No integrated learning / journaling tied to trading activity.

The User Pain Point

PRODUCTIVITY LOSS

Switching tabs creates context loss & decision fatigue.

DATA OVERLOAD

Raw data without smart filtering → analysis paralysis.

Proposed Solution

Core Components



Single Backend API

A high-performance FastAPI backend that aggregates all stock analysis tools into a unified interface.



Intelligent AI Agent

LLM-powered chatbot accessed via WebSocket that understands user intent and autonomously executes analysis tools.



Real-Time Watchlists

Dynamic watchlist management with live price tracking powered by yfinance data integration.



Automated "Best Predictions"

A complete workflow that scrapes Chartink, runs technical analysis, filters stocks, generates PDF reports, and emails them.

Architecture Benefits



Modular Tool Design

Plug-and-play architecture allows new analysis tools to be added without modifying the core agent logic.



Frontend Agnostic

Clean REST API + WebSocket design ensures seamless connection to any web or mobile frontend interface.

Automated Pipeline



SCRAPE



ANALYZE



REPORT



EMAIL

Technology Stack

Backend & DB

 **FastAPI**
Python Web Framework

 **PostgreSQL**
Relational Database

 **Uvicorn**
ASGI Server

 **WebSocket**
Real-time Comms

AI & LLM

 **LangChain**
Orchestration Framework

 **Groq API**
Llama 3.3 70B Model

 **ChromaDB**
Vector Database

 **Embeddings**
Sentence Transformers

Financial Data

 **yfinance**
Stock Market Data

 **TA-Lib**
Technical Analysis Lib

 **pandas-ta**
Data Manipulation

Automation

 **Playwright**
Scraping & PDF

 **BeautifulSoup**
HTML Parsing

 **SMTP**
Email Delivery

CORE CAPABILITIES

Key Modules & Features



01

User Authentication

Secure registration and login functionality using email and password via robust REST APIs.



02

AI Chatbot Agent

WebSocket-based conversational AI that understands natural language to execute analysis tools.



03

Watchlist Manager

Create/delete watchlists, manage stocks, and track live prices with yfinance integration.



04

Technical Engine

Calculates MACD, RSI, ADX, trend direction, and identifies chart patterns automatically.



05

Fundamental Analysis

Analyzes P/E, ROE, debt, and 30+ metrics to generate composite fundamental scores.



06

News Analysis

Scrapes stock-specific news and Pulse updates, providing AI-driven sentiment context.



07

Chart Capture

Uses Playwright to automatically capture and deliver candlestick chart images for visual analysis.



08

Prediction Workflow

Automated pipeline: Scrape → Analyze → Filter → Report → Email using LangGraph.



09

Nifty Comparison

Benchmarks user watchlist performance against Nifty index movements.



10

Learning Journal

Digital journal for users to log daily trading learnings, sentiments, and exceptions.



11

Conversation History

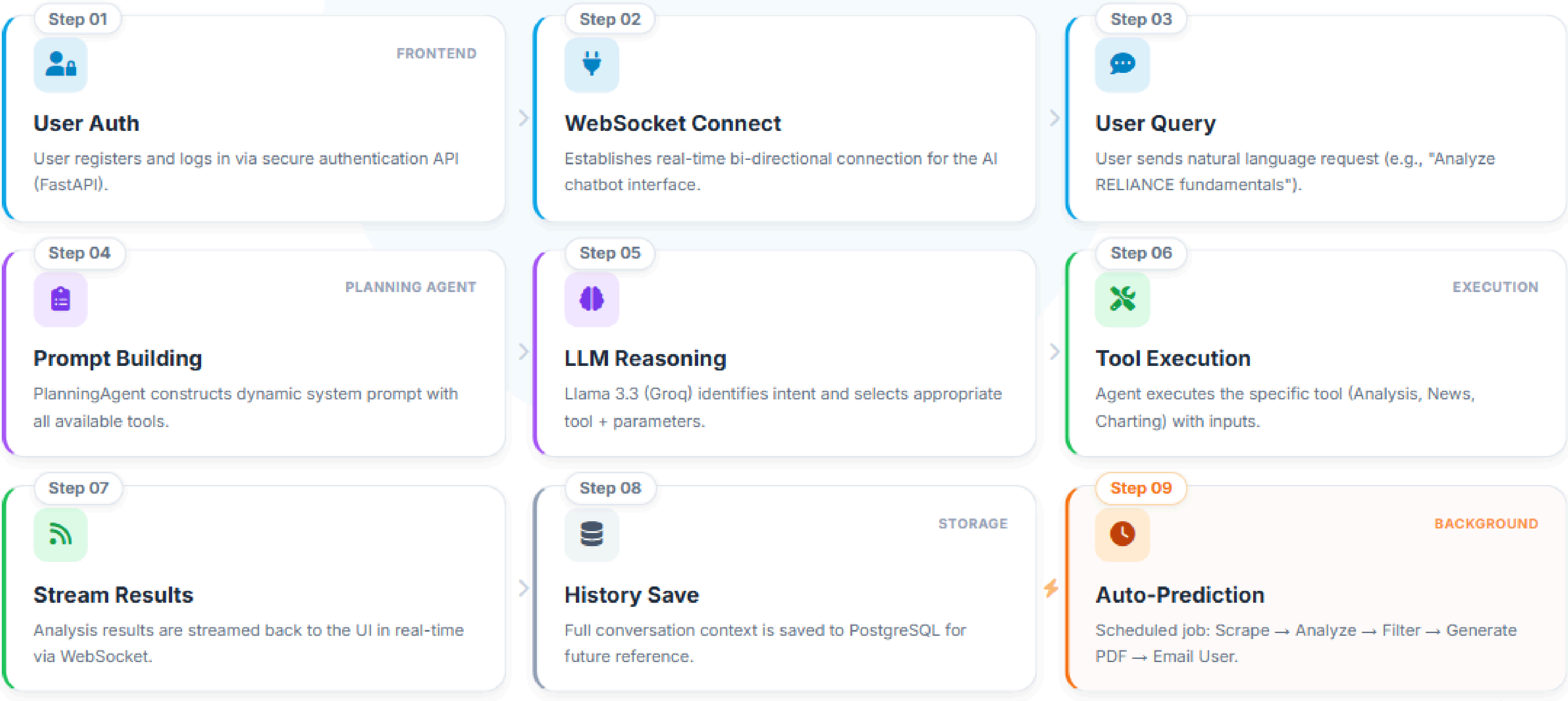
Maintains full chat history per user/session with complete CRUD operations.



More Coming Soon

How It Works

User Interface AI Agent Execution



AI Agent — The Brain

The PlanningAgent



LLM-Powered Engine

LangChain + Groq Llama 3.3 70B, high-speed reasoning, minimal latency.



Contextual Memory

Maintains full conversation history across interactions.



13+ Specialized Tools

Fundamental analysis, technical indicators, news, watchlist management.

LangChain

Groq LPU

Llama 3.3 70B

Decision & Execution Logic



Dynamic Prompting

Agent constructs system prompt listing tools based on user request.



Autonomous Routing

LLM decides tool & arguments, or answers conversationally.



Async Streaming

Real-time feedback over WebSocket via async generators.



Modular Architecture

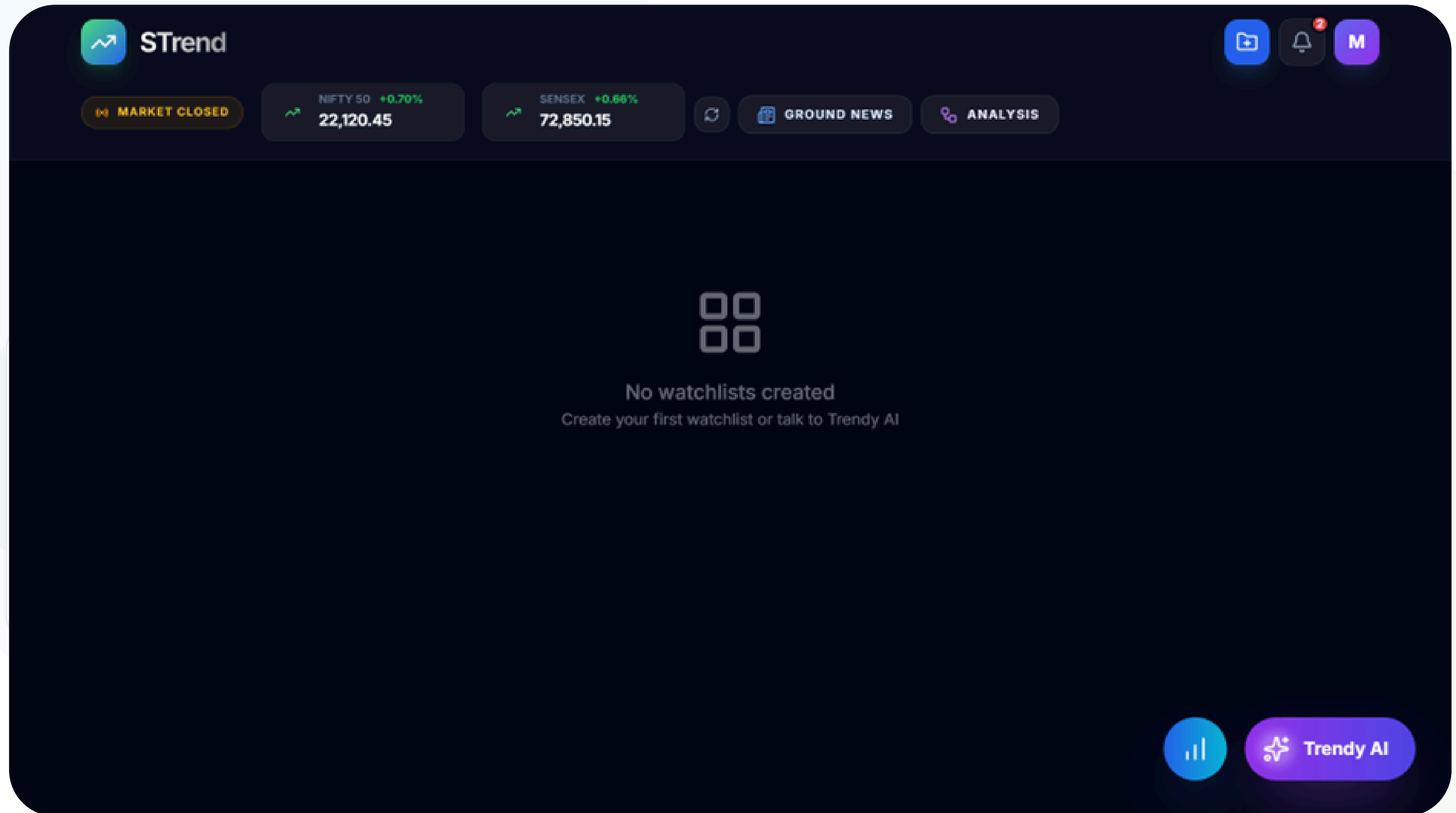
Scalable design: create a new tool file, register it, agent learns new skill instantly.

```
tools/technical.py
```

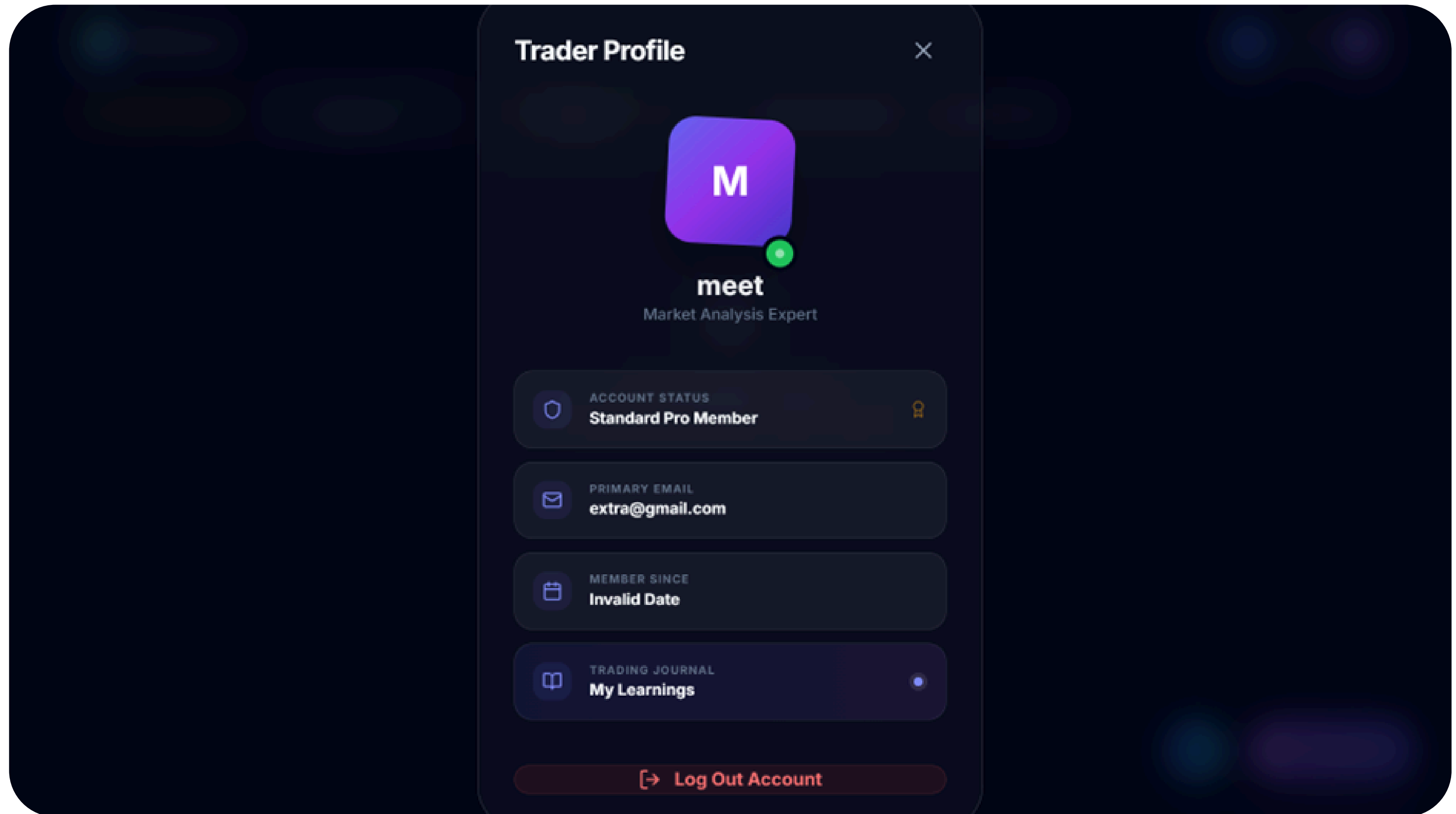


```
agent_registry
```

UI - Homepage



UI - ProfileCard



UI - Watchlist

watch list1

Analyze

↺

↻

+ Add Stocks

MARKET CLOSED

NIFTY 50

+0.70%

22,120.45

SENSEX

+0.66%

72,850.15

↺

GROUND NEWS

ANALYSIS

March 2026

1 stocks

TCS

Tata Consultancy Servi...

₹2,398.8

↗ 15.00 (0.63%)

ADDED DATE

24 Mar 2026

ADDED AT

₹2,398.8

TOTAL RETURNS

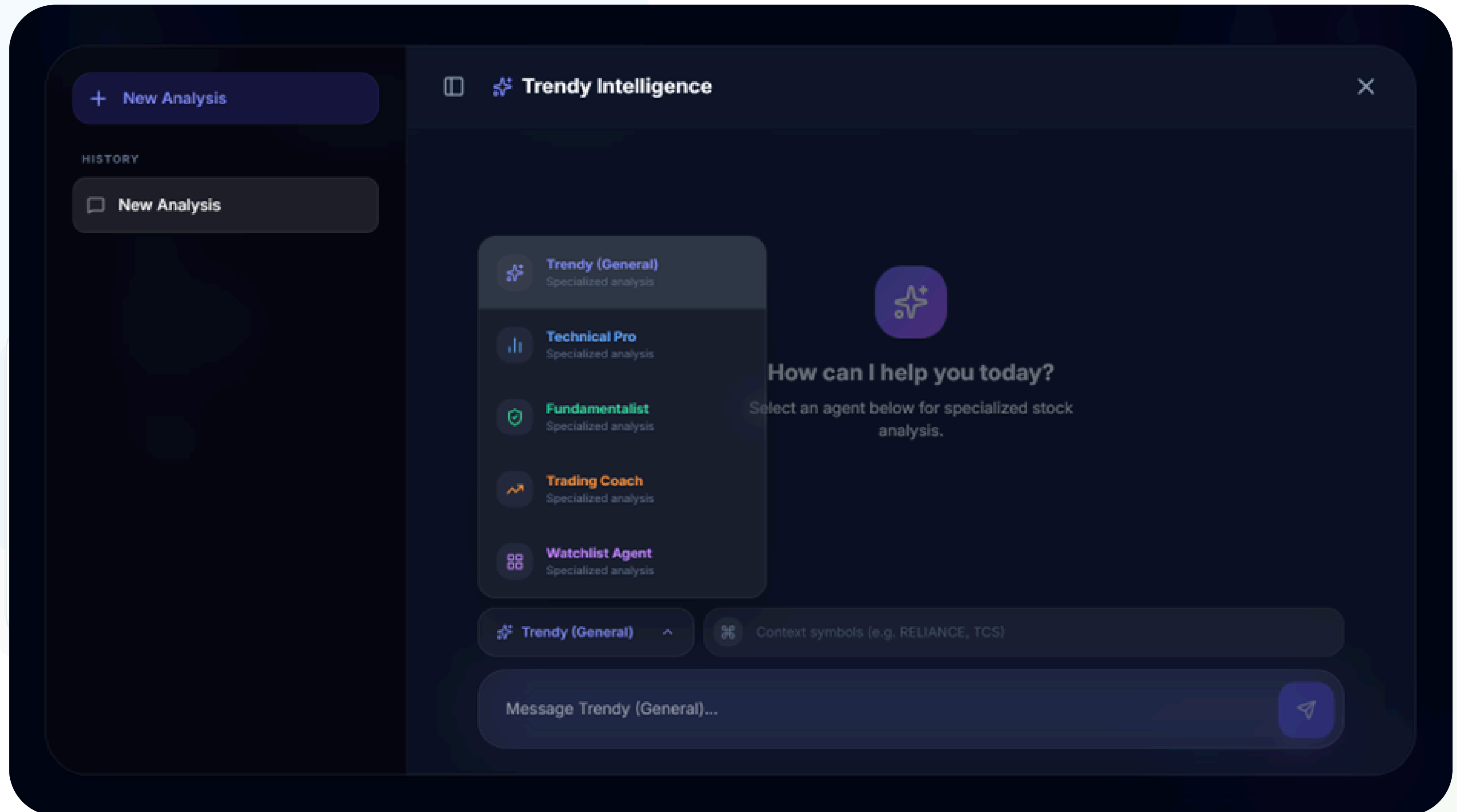
+0.00%

Trendy AI

UI - Charts



UI - AI Screen



Challenges Faced

Problem

Solution



AI Engine

LLM Hallucination

❗ AI invoked non-existent tool names causing failures.

✅ SOLVED

Strict input validation + explicit error feedback loops forced self-correction.



Data Pipeline

Ticker Normalization

❗ Mismatch "RELIANCE" vs "RELIANCE.NS".

✅ SOLVED

Centralized normalization utility auto-appends exchange suffix.



System

Playwright on Windows

❗ Async Playwright crashed due to event loop incompatibility.

✅ SOLVED

WindowsProactorEventLoopPolicy configured for stable async execution.



WebSocket

Real-time Streaming

❗ Async generators vs coroutines complexity.

✅ SOLVED

Standardized tool outputs as async iterators + careful WebSocket type handling.



Scraping

Scraping Fragility

❗ HTML structure changes broke scrapers.

✅ SOLVED

Resilient CSS selectors + fallback parsing + automated health checks.



Database

Schema Evolution

❗ Rapid dev led to schema changes risking consistency.

✅ SOLVED

Migration strategy with version control to manage updates safely.

PRODUCT ROADMAP

Future Scope

■ Phase 1: Experience ■ Phase 2: Intelligence ■ Phase 3: Ecosystem



Frontend Development

SHORT TERM

Build a responsive React/Next.js interface with interactive charts, dashboards, and dedicated chat UI.



Portfolio Tracker

SHORT TERM

Enable manual trade entry to track actual portfolio performance, P&L, and asset allocation.



Mobile Application

SHORT TERM

Develop native apps using React Native or Flutter for on-the-go analysis and notifications.



Advanced AI (RAG)

MID TERM

Implement Retrieval Augmented Generation with ChromaDB to let the AI learn from historical analyses.



Multi-Agent System

MID TERM

Deploy specialized agents for distinct domains like Options, Mutual Funds, and Crypto analysis.



Push Notifications

MID TERM

Real-time alerts when watchlist stocks hit target prices or trigger specific technical indicators.



Social Features

LONG TERM

Allow users to share watchlists, learning journals, and analysis reports with the community.



Backtesting Engine

LONG TERM

Simulate trading strategies against historical data to validate performance before live trading.



Broker Integration

LONG TERM

Connect directly with Zerodha/Upstox APIs for seamless order placement from within Strend.

Conclusion

“

Strend successfully demonstrates how AI technology can transform stock market research from a time-consuming, fragmented manual process into a **simple conversational experience**.



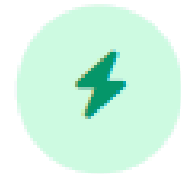
Unified Platform

Combines Technical, Fundamental, and News analysis in one view.



Intelligent AI

Context-aware chatbot that executes tools based on natural language.



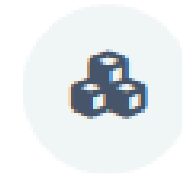
Real-Time Data

Live streaming of prices and analysis results via WebSocket.



Automated Workflows

Background agents for scraping, predicting, reporting, and emailing.



Modular Design

Scalable architecture allowing easy addition of new tools and features.



Empowering Faster, Smarter Trading Decisions.

Thank You!

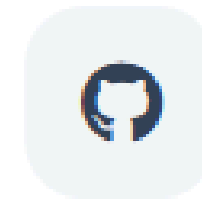
Any Questions?



GET IN TOUCH

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EXPLORE THE CODE

github.com/kevalmadariya/Strend



STREND

Smart Insights. Smarter Decisions.